



HSC Mathematics (QLD Specialist)

Matrix Practice

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Download PDF and resources:

[https://github.com/vuhung16au/
math-olympiad-ml](https://github.com/vuhung16au/math-olympiad-ml)



Read online at Vu's Maths Hub:

<https://vumaths.com/>

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*"As for everything else, so for a mathematical theory:
beauty can be perceived but not explained."*

Arthur Cayley (Founder of matrix algebra)

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1 Introduction

Remark 1.1 (Syllabus Note)

Matrices are not part of the current NSW NESA Mathematics syllabuses. The content in this booklet is adapted from the Queensland (QCAA) Specialist Mathematics syllabus. It serves as an excellent enrichment topic for students interested in linear algebra, computer graphics, and machine learning.

1.1 Project Overview

This booklet is a growing collection of matrix problems. It introduces matrix arithmetic, determinants, inverses, and applications to solving systems of linear equations and geometric transformations.

1.2 Target Audience

This resource is written for students who want to explore linear algebra beyond the standard NSW syllabus, in a clear, classroom-friendly format.

1.3 How to Use This Booklet

Read the short summary of the core formulas first, then attempt each problem before reading the solution.

1.4 Knowledge Needed

The key background knowledge is:

- Basic arithmetic and algebraic manipulation.
- Understanding of systems of linear equations.
- Basic 2D geometry (for geometric transformations).

1.5 Core Formulas and Ideas

2 Part 1: Worked Problems

2.1 Basic

2.2 Medium

2.3 Advanced

3 Part 2: Practice and Challenge Bank

3.1 Warm-Up Drills

3.2 Stretch Problems

3.3 Challenge Corner

4 Conclusion

Matrices provide a powerful language for describing and solving complex systems. As this booklet grows, it will become a compact reference for matrix operations and transformations.

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Repository: <https://github.com/vuhung16au/math-olympiad-ml/tree/main/HSC-Matrix>

A Appendices

A.1 Further Reading in Linear Algebra